

Learning Discriminative Representations to Interpret Image Recognition Models

Thèse de Doctorat

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Motivation

Low Stakes

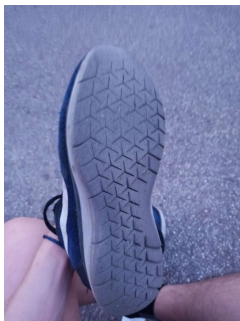
My go to exercise is running, **but...**

Motivation

Low Stakes

My go to exercise is running, **but...**

I think my running shoes
are getting *worn*

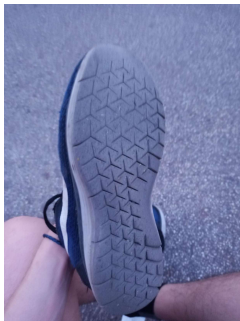


Motivation

Low Stakes

My go to exercise is running, **but...**

I think my running shoes
are getting *worn*



I want a replacement,
but I know about
machines, not shoes!



Motivation

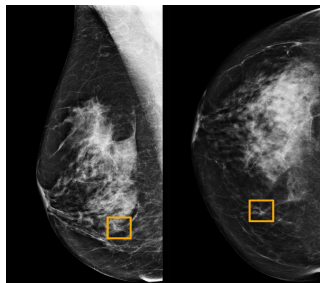
Raising the stakes

Now let's consider relevant situations:

Motivation

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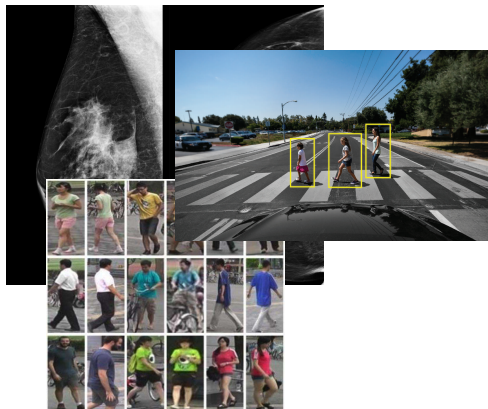
Now let's consider relevant situations:



Motivation

Raising the stakes

Now let's consider relevant situations:



Straight to the point



- How do we **know how** a system works?
- How do we **know how** safe a system is?
- If a system fails, **who** is accountable?

We must **understand** the
behaviour of these models.

Step by step

Computation,
Computer
Vision and AI



Explainable AI



Thesis
objectives

Computation, Computer Vision and AI

Computation



Better known as *Computer Science*.

Study of:

- Algorithms.
- Data structures.
- Design of hardware and software.

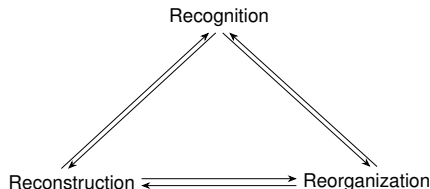
Computation, Computer Vision and AI

Computer Vision

Replication of human vision capabilities.

Three fundamental tasks[1]:

- Recognition.
- Reconstruction.
- Reorganization.



Computation, Computer Vision and AI

Artificial Intelligence

Systems capable of performing tasks requiring human intelligence [2].

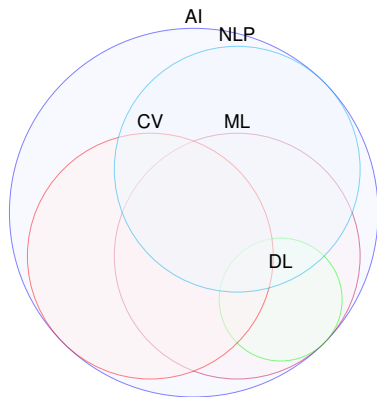
Computation, Computer Vision and AI

Artificial Intelligence

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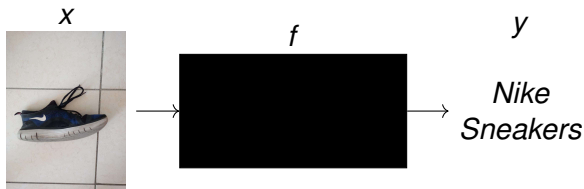
Subfields:

- Machine Learning (*ML*) & Deep Learning (*DL*).
- Computer Vision(*CV*).
- Natural Language Processing (*NLP*).
- Robotics.



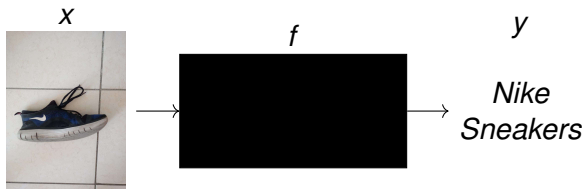
Explainable AI

We are interested in understanding models, behaving like a black box model:



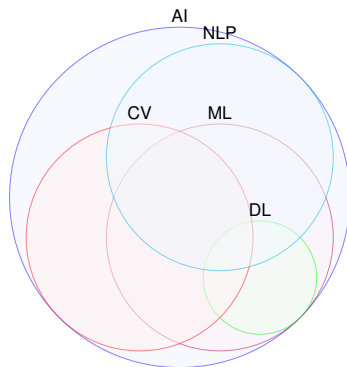
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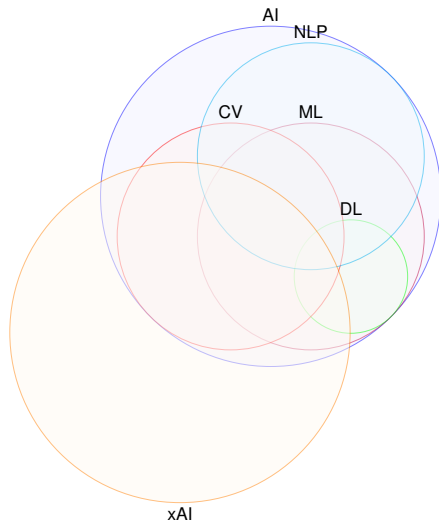


We want to *know why* $f(x) \rightarrow y$

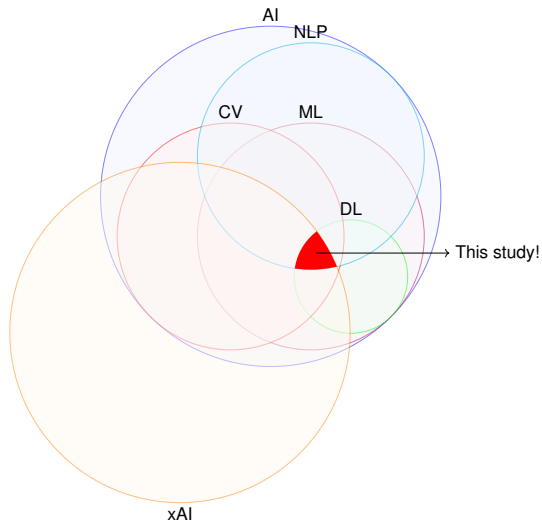
Fitting it all together



Fitting it all together



Fitting it all together



Thesis Objectives

Improvement of recognition and interpretable properties of model predictions.

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Improvement of recognition and interpretable properties of model predictions.

In particular:

- Development of low cost/complexity explainability approaches.
- Establishment of a fixed evaluation protocol.
- Differentiation of human based and machine explanations.

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Preliminaries

Desired data of Interpretability Study [4]

We ask questions regarding
black box models.

- How does it work?
- How safe is it?
- Who is accountable for it?
- Who benefits from it?
- Who uses it?

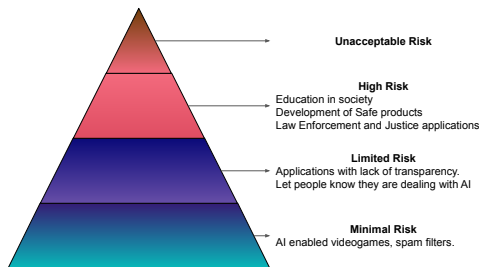
**Accepted AI systems within
society must answer this.**

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Regulation planned with the European AI act[3]

Accepted AI systems within society must answer this.

Image Recognition Models

Convolutional Neural Networks (CNN)

Based on the **convolution operation**.
A representation $f \star g$ is computed for a feature map f and a kernel g .

First approach with *Neocognitron* [5]

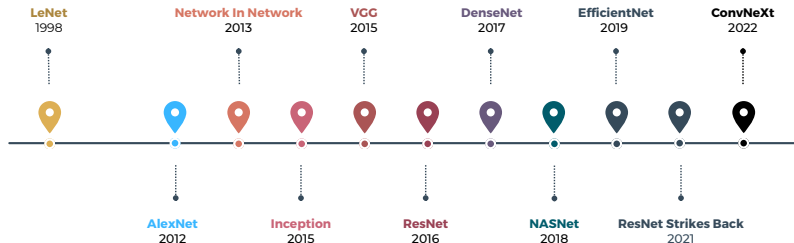
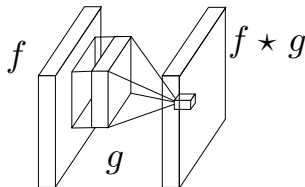


Image Recognition Models

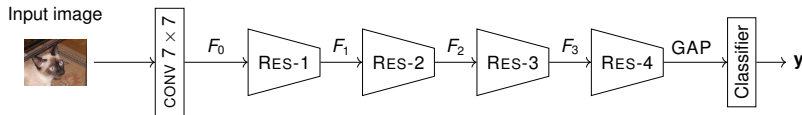
Convolutional Neural Networks(CNN)

Training deep models is difficult
Gradient information can be lost with depth.

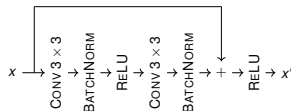
Image Recognition Models

Convolutional Neural Networks(CNN)

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Gradient information can be lost with depth.
A model surviving the test of time: *ResNet* [6].



Basic Block



Bottleneck Block

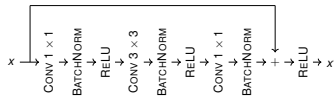


Image Recognition Models

Self Attention Architectures

Transformers

Attention: compute the relevance of each element relative to others in an embedding.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{Q \times K^T}{\sqrt{d_k}}\right) V$$

Image Recognition Models

Self Attention Architectures

A modification of transformers to vision, the *Vision Transformer* (ViT) [8]

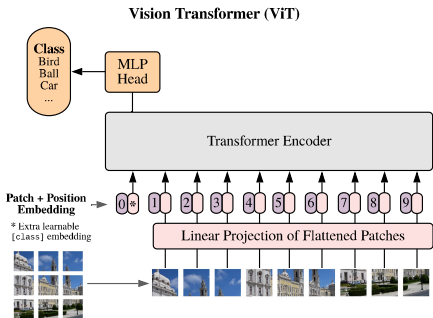


Image Recognition Models

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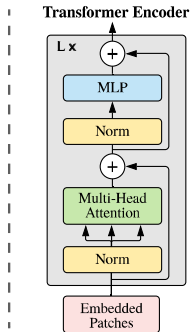
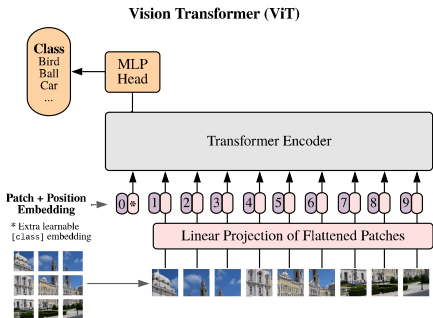
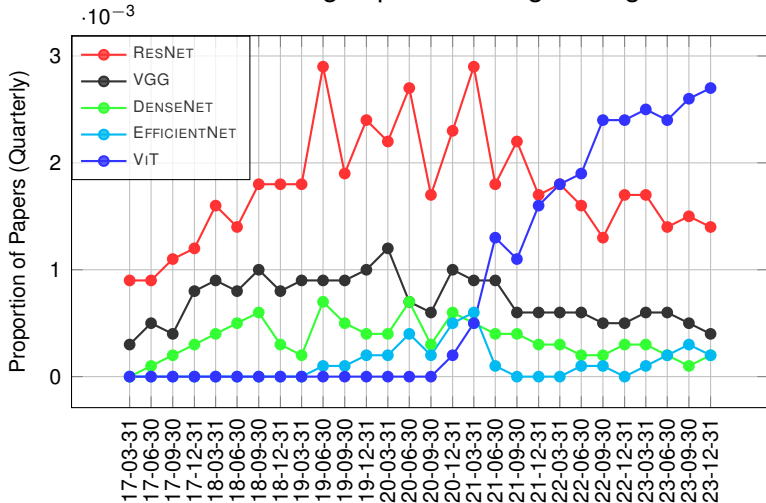


Image Recognition Models

The Current Landscape

Transformers had a strong impact on image recognition:



Interpretability

Two dimension grouping

Transparency

Approaches requiring changes to an architecture or training process.

Post-Hoc Interpretability

Approaches utilizing information contained within a model.

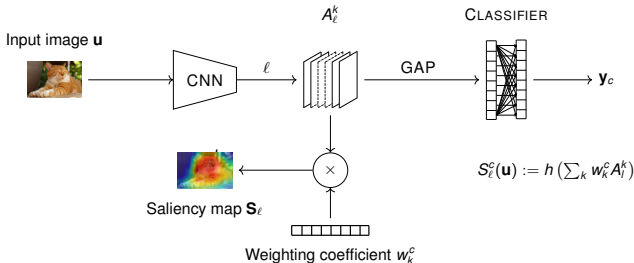
Interpretability

Class Activation Methods

Semantic information is found in deeper stages of a model [9].



Salient regions are highlighted for an input image:



Interpretability

Evaluating Interpretability

Saliency: what is relevant towards a prediction.
An explanation should demonstrate similar predictive properties to its query.

Input image. (I)



$$Y_i^c = 0.8756$$

Explanation Map. (E^c)



$$O_i^c = 0.7442$$

Interpretability

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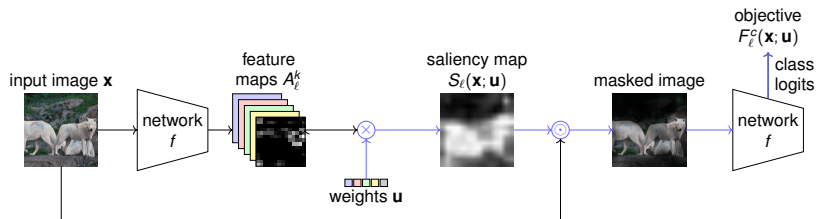
$$O_i^c = 0.7442$$

- Prediction loss when using an explanation. (Average Drop[10])
- Ratio of explanations being better than the input image. (Average Increase [10])
- Reaction to saliency guided modifications (Insertion/Deletion [11])

Preliminaries

Premise

Is it possible to optimize the weighting coefficient w_k^c to obtain a saliency map with the highest probability of prediction?



Preliminaries

Contributions

Opti-CAM

Maximization of predicted probability of an explanation.

Average Drop

Evaluation metric designed to enhance comparisons.

Opti-CAM

Motivation

● Combination of Feature Maps

To compute w_k^c , Score-CAM [12] considers feature maps independently.
What about a linear combination?

$$F(\mathbf{w}) := f \left(\mathbf{u} \odot n \left(\text{up} \left(\sum_k w_k A_l^k \right) \right) \right)_c$$

F is optimizable with respect to $\mathbf{w}$ and an initial w_0 via backpropagation.

Opti-CAM

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● Masking-based Saliency

Saliency maps produced by optimization in the input space [13, 14]

The optimization usually takes form of:

$$S^c(\mathbf{u}) := \arg \max_{\mathbf{m} \in M} f(\mathbf{u} \odot n(\text{up}(\mathbf{m})))_c + \delta R(\mathbf{m})$$

Opti-CAM

Optimization

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